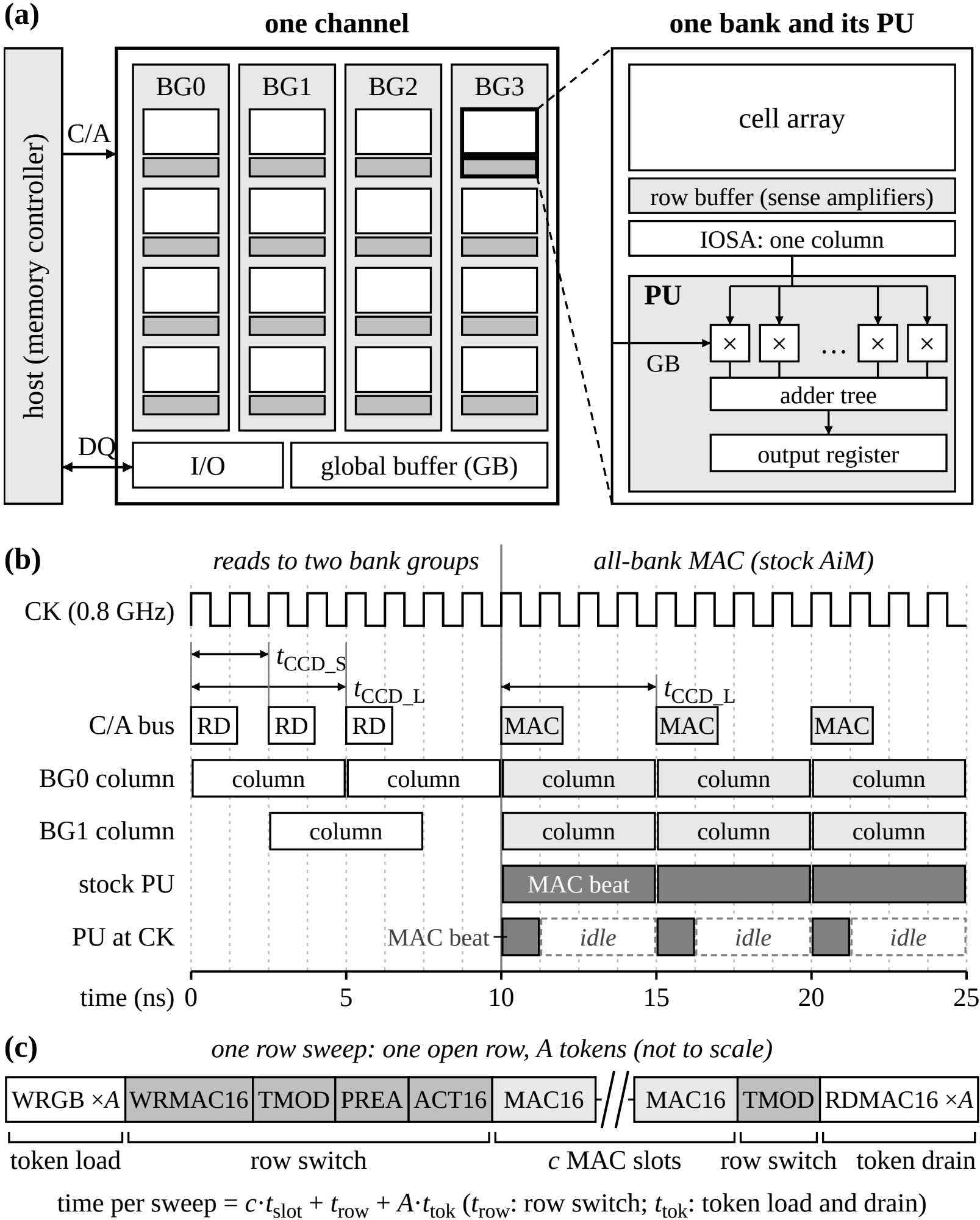
(a)
one channel
one bank and its PU
host (memory controller)
C/A
DQ
BG0
BG1
BG2
BG3
I/O
global buffer (GB)
cell array
row buffer (sense amplifiers)
IOSA: one column
PU
GB
adder tree
output register
(b)
reads to two bank groups
all-bank MAC (stock AiM)
CK (0.8 GHz)
t_{CCD_S}
t_{CCD_L}
t_{CCD_L}
C/A bus
RD
RD
RD
MAC
MAC
MAC
BG0 column
BG1 column
column
stock PU
MAC beat
PU at CK
MAC beat
idle
idle
idle
time (ns)
0
5
10
15
20
25
(c)
one row sweep: one open row, A tokens (not to scale)
WRGB ×A
WRMAC16
TMOD
PREA
ACT16
MAC16
MAC16
TMOD
RDMAC16 ×A
token load
row switch
c MAC slots
row switch
token drain
time per sweep = $c \cdot t_{slot} + t_{row} + A \cdot t_{tok}$ (t_{row}: row switch; t_{tok}: token load and drain)